Phrasal Provenance: An LLM-Assisted Methodology for Elite Network Research

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Abstract

We present a methodology for LLM-assisted research that anchors every claim to a specific sentence in a source document. The paper situates this methodological intervention within the diffuse field of elite research and the contested question of where power lies in the European Union. Using European institutional elites as a case study—commissioners, DGs, and MEP leaders across seven commission terms (1995–2029)—we demonstrate a pipeline that scrapes from controlled sources, normalises data into a relational database with mandatory provenance links, uses LLMs for phrase-level fact extraction with JSON manifests, and cross-validates every claim against source text to detect hallucinations. The system surfaces substantive findings including the collapse of the French *grande école* pipeline into the Commission and a stark asymmetry in elite network ties between commissioners and MEP leaders (0 exclusive networks). We report on failure modes encountered—LLM hallucination and entity duplication from non-deterministic prompt outputs, and the "CFR is not ECFR" problem in string-similarity deduplication—and describe the methodological choices that address each.

1 Introduction

1.1 Elites in social theory

One of the signature features of the research agenda driven by structural-demographic theory has been its emphasis on elites, and how they figure in the dynamics leading to societal destabilisation, crisis and sometimes collapse. Needless to say, SDT researchers have not been the first to single out elites for social scientific attention. Indeed, there is little doubt that in one way or another, elites have represented an area of considerable interest for students of society and politics since the very dawn of this domain of human inquiry, as attested by Plato's argument for government by philosopher-kings and Aristotle's detailed commentaries on the relative merits and demerits of oligarchy, aristocracy and democracy.

However, in the era of modern social science, elites have been the subject of more dedicated and sustained attention since at least the time of the Italian elite theorists of the late nineteenth and early twentieth centuries—authors from whom we have inherited concepts such as the "organised minority" (Mosca), the "circulation of elites" (Pareto), and the "iron law of oligarchy" (Michels).

Subsequent generations of social scientists have looked at elites from somewhat different angles again. From the mid-twentieth century onwards, studies such as those of C. Wright Mills and G. William Domhoff asked searching questions about who truly governed, and about where the agency behind crucial decisions truly lay in polities where elected officials co-existed

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with powerful unelected (and often non-public-facing) elites. These arrangements were conceptualised in different ways: Mills emphasised the overlap between political, military and corporate elements in the "power elite" that shaped US governance [7]; Domhoff promoted a class-domination theory of power; Michael J. Glennon described the bifurcation between the outward-facing "Madisonian" institutions and the more obscure "Trumanite" network in terms of "double government"; while Peter Dale Scott spoke of "deep politics" or simply the "deep state". However, what these perspectives had in common was the idea that the reality of power looked very different from the pluralist models associated with leading political scientists such as Robert Dahl.

Meanwhile on a more sociological level, scholars such as Pierre Bourdieu and his many acolytes have focused on the processes (often mediated through elaborate cultural and symbolic codes) through which elites ensure social reproduction across generations.

1.2 Elites and the European project

More recently, political scientists working on both sides of the Atlantic have charted various routes through which political elites have seemed to increasingly insulate themselves from the cut and thrust of what used to be considered the basic operations of everyday democratic politics, leaving "cartel parties" and a hollowed out political "void" in their wake (Mair). Some of this research has been focused on the national level, while some of it has honed in on the role of supranational institutions as key sites in the distancing of political power from democratic checks and balances as traditionally understood. Some of it has charted the emergence of a new supranational European elite on the back of the institution-building processes, as Verzichelli and colleagues document [1]. Perhaps the best-documented example of this purported phenomenon is that of the European Union and its component organs, which has spawned an extensive scholarly literature and become something of a byword for democratic deficit—at least in the eyes of its critics, and sometimes, somewhat grudgingly, even in the eyes of its supporters. Much of this EU-oriented literature has been focused on the workings of the European institutions—sometimes casting them explicitly as venues where elite-level decision-making can be hived off from the democratic mechanisms notionally built into national domestic politics—.

This has dovetailed with recent research by a number of political scientists based in the US who have deployed fine-grained quantitative research tools to interrogate that polity's purported democratic credentials, only to find that US government policy appears to be shaped overwhelmingly by elite preferences, while the preferences of non-elites seem to exert negligible influence (Gilens and Page). Though not explicitly framed by the researchers in terms of "elite theory", their work is clearly pertinent in this regard.

Comparative work on European power elites likewise finds pronounced social immobility [2].

Taken together, all of this might seem to indicate the existence of a thriving ecosystem of elite research, whatever its implications for the socio-political health of contemporary societies. However, the relative abundance of research pertaining in one way or another to elites belies a remarkably diffuse field of enquiry. Across this landscape of elite-focused studies, we find research questions that are not just distinctly different from one another, but sometimes in certain important respects pulling in opposite directions. Some of these questions concern the perpetuation of elite dominance. Others, arguably, are about something close to its inverse. Even within the narrower field of specifically SDT-driven research, we find distinctly different kinds of questions about elites (and would-be elites) being formulated. Some of these are concerned with what elites try to do (often unsuccessfully) to stave off crisis and collapse in tumultuous times. Others are concerned with how the overproduction of elites or elite aspirants directly *contributes* to societal destabilisation. This latter seam of research can be split, in turn, into at least two rather distinct substrata. To put it somewhat crudely, this means looking not just at elites as "winners", but at elites (or perhaps more accurately "would-be", "wannabe" or "has-been" elites) as "losers". Both phenomena can be highly destabilising. As Turchin

has pointed out, the overproduction of the elites that we might call "winners" tends to result in the ratcheting up of the "wealth pump", leading to unsustainable levels of extraction from the wider economy and society, and the gradual impoverishment of non-elites. Meanwhile, the overproduction of "elites as losers" poses its own problems, which can take a number of forms. One example might be that of the *ancien régime* French nobles who lashed out against the crown's attempted reforms in the run-up to the crisis of 1789, seeing them as only being likely to speed up the trajectory of downward social mobility they perceived themselves to be on, thereby unwittingly undermining the remaining supports of the system that they clung to, limpet-like, for sustenance, and ultimately accelerating France's slide into revolution and civil war. Another example might be the heavily indebted graduates of US law schools and other advanced degree programmes who are currently trying and failing to turn their credentials into lucrative careers, and increasingly breaking away from the traditional political formations to support more radical offerings—identified by Turchin and other SDT researchers as a recipe for serious instability.

Thus "elite research" is not one thing, but many things, and any examination of this shorthand immediately throws up fundamental questions, including but not limited to the following: Whom are we really talking about when we talk about "elites"? How many different types of "elite" are there, and who should really be called "elite"? In so far as "elite" is a meaningful concept, how does one attain this status, or maintain it in an era in which many individuals with "elite" status are experiencing downward social mobility and becoming *déclassés*? Relatedly, how does one lose elite status? Do traditional means of attaining or maintaining elite status, such as acquisition of higher education (cf the famous "college premium"), still apply, or is this now being washed away in a sea of credentialism, degree inflation and unsustainable student debt? What is the relation between the "winner" and "loser" elites? Is there a firm distinction between the two, or more of a revolving door?

And perhaps most importantly: how do these various questions relate to the moment of profound and multifaceted crisis that Western societies are currently experiencing?

1.3 Why the European Commission?

To be sure, this represents a tangled skein of questions rather than a strict research agenda, and while we cannot address all of them in our paper, we do propose to at least bring together a number of distinct strands of elite research as they pertain to Europe's current conjuncture. In so doing, we hope to achieve a number of goals. As well as bridging the gap between these somewhat disparate strands of elite theory, we aim to apply the resultant framework to relatively novel subject matter. Previously, SDT-inspired elite research has been applied to various historical and contemporary societies, but never, to our knowledge, to a supranational entity or supranational elite. Something similar could be said of "power elite" research along the lines of Mills or Domhoff. No consensus may exist as to how to explain what happens in the inner sanctums of American power, but at least we have some reasonably firm hypotheses to choose from, and to build upon. How power works at the level of Europe, meanwhile, is notoriously something of a black box.

This is not to draw an equivalence between American and European power. While the EU has long been recognised as a behemoth in areas such as trade, scholars have continuously debated the extent to which it can be considered a coherent political actor in a larger sense. Indeed, some scholars argue that, insofar as the EU is a power on the international stage, it is a declining one. Nonetheless, declining or otherwise, one cannot attempt to make sense of the current conjuncture without attempting to make sense of the EU and the elites that animate it. Whatever one thinks about the efficacy of the EU, it is no longer possible to think about German or French policy, either on a domestic or foreign level, in isolation from European policy. Nor is it possible to think through the reactivation of Europe as a major theatre in the current not-so-cold war between East and West without considering the EU's role.

Of course, part of the reason that the inner workings of the EU are so obscure is that the institutional architecture of the European project is unusually complex. "Studying the EU" could mean many different things, and the first questions one faces when attempting to do so are "how?" and "where to begin?". Among the various institutions that comprise the EU, we have chosen to focus our initial analysis on the European Commission, as we believe that if any one entity embodies the supranational elite that is emerging on the back of the European project, it is this more or less permanent central bureaucracy. (Not for nothing is the Commission sometimes described as the beating heart of the EU.) However, we acknowledge that this basic starting premise of our analysis is open to a certain degree of debate, insofar as no consensus exists in the scholarly literature as to where power lies within the European project, or even as to the European project's basic nature. These questions have been the subject of many decades of intense debate, and while we cannot make them the focus of this paper, we do need to briefly address their basic outlines, insofar as they pertain to the current analysis.

Put simply, the various schools of thought on the nature of the European project, while manifold, can roughly be divided into two broad camps—that which understands the EU as a fundamentally inter-governmental initiative, and that which views it as an experiment in supranational governance. Furthermore, this difference in perspective is not merely scholarly or theoretical: the EU clearly incorporates both inter-governmental and supranational elements and, what is more, is riven by tensions over this very issue. That is, the theoretical difference in opinion maps onto a real-life one, in that the protagonists of the European project have themselves chronically disagreed over whether it ought primarily to be an intergovernmental or supranational enterprise. And while the actually existing structures incorporate both logics, it is difficult to say conclusively which is more important, as those structures have evolved on the back of multiple compromises, via multiple rounds of treaty negotiations, between actors with rivalling visions as to which logic ought to prevail. Thus this tension is woven into the basic fabric of the European institutions, and snapshots taken at different historical junctures since the founding of the European institutions would show the battle lines drawn in different ways between the intergovernmentalists and the supranationalists, with the initiative periodically slipping back and forth between the two dispositions.

In short, the question of where power lies in Europe is shrouded in some opacity, and we do not need to reach for the scholarly literature to see evidence of this. It is reflected in many of the stock phrases customarily used in real life to describe the workings of the European project. Indeed, when facing criticisms from "Eurosceptic" political actors who lament the eclipse of national sovereignty, their "pro-European" counterparts cheerfully acknowledge that the EU is built on a principle of "pooled sovereignty". But it is not always clear where exactly this sovereignty "pools together", as it were. Indeed, an oft-repeated line widely attributed to Henry Kissinger reflects this opacity. In reference to the diffuse nature of the European power structure, Kissinger is purported to have quipped: "Who do I call when I want to speak to Europe?"

On the other hand, some commentators would suggest that, if US leaders ever really did struggle to "speak to Europe", as Kissinger is said to have suggested, this issue would seem to have been less salient in recent times. From this point of view, Europe has, in spite of increased trans-Atlantic verbal sparring in the Trump era, largely abandoned the independentist streak of the 1990s and early 2000s and increasingly fallen into line with the United States, at least on key foreign policy issues.

To the extent that this is true—and clearly the point is controversial—it raises a further difficult question for anyone attempting to carry out research on the EU elite, and another dimension to the question of where power lies. That is, we are no longer merely pondering whether power really lies in the Commission or in the Council, or in Brussels, Frankfurt, or the various national capitals. We also must take seriously the question of whether it even lies in Europe.

In sum, the study of "elites" is the study of many different things: rising elites, declining elites, would-be elites, entrenched elites, "power elites" and "elites" who may not yet be obviously experiencing downward social mobility, but whose real power is open to question. In turning our focus to the cadres of the European Commission, we leave open the question of exactly what kind of elite it is. However, taking it that it is an elite of some kind, and that shedding light on how it is constituted would be useful in the current conjuncture, we take a preliminary step toward this goal in the following sections.

1.4 From elite theory to a citation-anchored method

The step we take is methodological. The evidentiary challenge follows from the conceptual one. Elite formation is distributed across 27 member states with distinct educational systems, career pathways, and political cultures; the networks that connect these elites cross national boundaries and institutional sectors [3]; and the relevant evidence is scattered across biographies, CVs, declarations of interest, and institutional records. Capturing it requires aggregating data across languages, jurisdictions, and source types, while keeping every claim traceable to its source.

The remainder of this paper describes a methodology that treats the LLM as an extraction engine, not an authority. Every claim in our database is linked to a specific sentence in a specific document. The LLM's output is a hypothesis, verified against the source before acceptance. The system currently holds several thousand claims across several thousand entities with sentence-level provenance, supporting analysis of seven European Commissions from Santer (1995) through Von der Leyen II (2024–2029).

2 Methodology

LLMs hallucinate because they generate text by sampling from a learned probability distribution over tokens. When the model lacks grounding in external facts, this distribution can produce plausible-sounding but incorrect statements [4]. Manakul et al. (2023) demonstrate that hallucinated outputs are detectable through self-consistency: when an LLM is asked the same question multiple times, hallucinated answers tend to diverge and contradict each other, while factual answers remain consistent across samples [6]. This reveals that hallucination is a property of *unconstrained* generation: the model's internal distribution over tokens is underdetermined by the prompt, and stochastic variance produces divergent outputs.

Crucially, Kadavath et al. (2022) show that LLMs are substantially better at *evaluating* their own outputs than at *generating* correct ones—they can detect errors in existing text at much higher rates than they can avoid producing errors during generation [5]. This generation-verification gap is the foundation of our approach: we do not ask the LLM to invent facts. We ask it to cite a numbered phrase from a source document that supports a factual claim. The task changes from "produce text from your internal distribution" to "does this specific sentence support this specific claim?"—a form of natural language inference with dramatically lower hallucination rates.

LongCite (2024) provides direct evidence that LLMs can reliably anchor claims to specific source sentences when trained and prompted to do so [8]. Our methodology extends this insight: by numbering phrases and requiring the LLM to return a phrase index for each extracted fact, we build a system in which every claim in the database is traceable to a specific sentence in a source document. The LLM acts not as a knowledge base but as a structured extraction engine with mandatory provenance.

2.1 System Architecture

The pipeline has five stages: scrape, extract, verify, resolve, and render.

2.2 Database Schema

The database implements a graph-style ontology on a standard relational backend. Every piece of data is stored as a triple: an *entity* (person, organisation, institution, commission, or source document) connected to a *property value* by a *predicate* (`educated_at`, `affiliated_with`, `served_on_commission`, `held_portfolio`, `from_country`). The property value may be another entity (forming a graph edge) or a literal value (a date, a degree, a country name).

The predicates are drawn from the substantive concerns of elite research: education (`educated_at`, `studied_field`, `held_degree`) to track elite formation; organisational ties (`affiliated_with`, `member_of`) to map network membership; career trajectories (`served_on_commission`, `held_portfolio`, `from_country`); and institutional position (`served_on_ep`). Each fact—each triple—carries mandatory provenance: the source document from which it was extracted, the exact sentence (stored as `quote_text`), and its position within that document (`phrase_index`). This transforms the database from a collection of assertions into an auditable knowledge graph: every edge can be traced to a specific sentence in a specific source. The four core tables are:

- `entity`: All nodes in the graph—persons, organisations, educational institutions, commissions, and source documents—with a type, category, and optional country code.
- `fact`: Directed edges connecting entities (`educated_at` links a person to a university) or attaching literal properties (`held_degree` links a person to "PhD"). Includes confidence and qualifier fields.
- `citation`: Metadata for each source document—URL, access date, source type (Wikipedia, Wikidata, Commission CV PDF, etc.).
- `provenance`: Every fact's audit trail—the `citation_id`, the `phrase_index` (sentence number), and the exact `quote_text` that supports the claim.

Facts without provenance are deleted rather than retained. This is not a cleanup operation but a structural invariant: the knowledge graph is self-auditing.

2.3 Phrase-Level Extraction

The core innovation is phrase-indexed extraction. Rather than sending an article to an LLM and accepting unstructured output, we:

1. Split source text into numbered sentences (50–80 per document).
2. Present the LLM with numbered phrases and a structured prompt: "Extract every organisation that X has been a member of. For each: phrase number, organisation name, role, reasoning."
3. Parse the JSON response into a *manifest*—a durable intermediate artifact stored as JSON.
4. Cross-check: retrieve the sentence at the claimed phrase index and verify it contains the asserted information.

The manifest preserves the LLM's full output including reasoning and phrase indices, making extraction auditable and reproducible.

2.4 Failure Modes

We encountered three categories of failure:

2.5 Methodology Failure: the Short-Extract Problem

Our initial extraction runs used Wikipedia API extracts, which returned truncated texts of 300–400 characters for many articles. The LLM, given insufficient context, hallucinated freely: it extracted "Dacian Julien Cioloş" (the commissioner's own full name) as an educational institution, and returned generic names such as "aeronautical engineer" as institution names. When we re-scraped the same biographies using raw HTML to obtain full texts (5,000–30,000 characters), the same LLM with the same prompt produced clean extractions with specific institution names and correct phrase indices. The problem was not the LLM's capability but the quality of the input: truncated source texts forced the model into a high-hallucination generative regime. Full-text input, combined with phrase-numbering and mandatory citation, moved the task into a low-hallucination verification regime.

2.6 Non-Deterministic Extraction

Running identical prompts on the same source text produced different sets of extracted organisations on different runs. The LLM's non-determinism meant that manifest regeneration changed the database. This is tolerable for additive processes (new facts are added, existing ones persist) but problematic for analytical reproducibility. We addressed this by storing manifests in version control as the canonical extraction record.

2.7 Entity Resolution (the "CFR is not ECFR" problem)

Each extraction produces entity names as they appear in source texts: "Radio Netherlands," "Radio Netherlands Worldwide," "FEPS," "Foundation for European Progressive Studies." These raw names carry their evidentiary basis—the phrase from which they were extracted and the LLM's reasoning about why that phrase indicates an organisation. Entity resolution proceeds by rolling these raw names into a canonical set: when two entities should be treated as identical, their facts are merged under a single canonical name.

String similarity alone is dangerous for this task. The sequence "CFR" vs "ECFR" scores 0.857 on difflib—above many reasonable thresholds—yet one is the US Council on Foreign Relations and the other is the European Council on Foreign Relations. Pure auto-merge at any threshold below 0.95 would incorrectly collapse them.

Our two-pass approach first auto-merges at string similarity ≥ 0.95 (safe: typos, punctuation variants). For borderline pairs (0.30–0.95), an LLM judge receives both raw names *plus* up to three evidence phrases from each side with the extracted role and reasoning. The LLM evaluates whether the two entities refer to the same organisation based on the full evidentiary context, not just the name. It correctly distinguishes CFR and ECFR (US vs European foreign policy evidence), while merging "Radio Netherlands" with "Radio Netherlands Worldwide" (rename detected from career context) and "FEPS" with "Foundation for European Progressive Studies" (French/English acronym expansion). Every decision is logged to `_dedup.json` with similarity scores, evidence excerpts, verdict, and reasoning—making entity resolution itself auditable and reproducible. The same methodology applies to educational institution names, merging variants such as "University of Vienna/Austria" into "University of Vienna" and Oxford colleges into "University of Oxford."

2.8 LLM-Based Organisation Classification

The same evidence-driven LLM judging approach extends to classification. With 800+ organisations in the database, we classify each into one of six types: `think-tank`, `political-party`, `corporate`, `government-body`, `industry-association`, or `ngo`. The LLM receives the organisation's name plus up to five connected people with their roles and categories as evidence, and returns a single classification. Each decision is stored as a `classified_as` fact with provenance

recording the evidence on which it was based. The classification is therefore auditable and regenerable. In the Obsidian wiki, these tags drive automatic colourisation of the graph view. The pipeline has classified over 1,000 organisations.

3 Wiki Output and Human Review

The database is rendered as a static wiki with 1,400+ interlinked pages using a Python generator (`genwiki.py`). Each page displays all facts about an entity with inline citation footnotes: a quoted sentence from the source document, a link to the source, and the exact source file path and character position. This makes every claim independently verifiable by a human reader in seconds—the reader sees the claim, the evidence sentence, and can navigate directly to the source file.

The wiki can be opened as an Obsidian vault, enabling graph-view navigation (Figures 1–2). Obsidian's graph view renders the 1,400+ entities as nodes and the 4,000+ facts as edges, automatically colourised by elite type using YAML frontmatter tags generated by the pipeline. Political elites (commissioners, MEPs) appear in blue, economic elites (corporate leaders) in red, administrative elites (DGs) in green, and judicial elites (CJEU) in teal. Organisations are further colour-coded by type: think tanks in grey, corporate entities in pink, and political parties in purple.

The graph view surfaces structural patterns that would be invisible in tabular data. Commissioners cluster densely around a small number of policy institutes (Bilderberg Group, Trilateral Commission, ECFR), while MEP leaders form a separate network centred on political party structures. Corporate elites occupy yet a third cluster around industry associations and financial institutions. The few bridging individuals—Jean Lemierre, Oliver Bäte—appear as connections between otherwise disconnected network components.

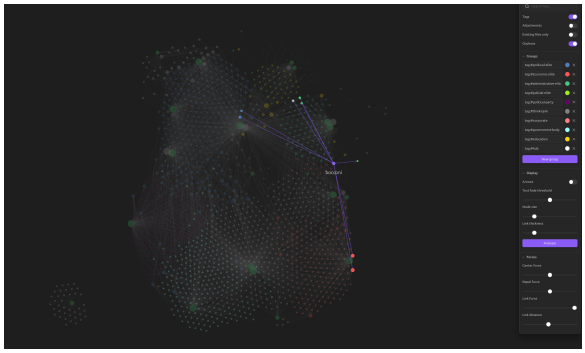


Figure 1: Obsidian graph view of the Euro-SDT wiki. Nodes colourised by elite type: blue (political), red (economic), green (administrative).

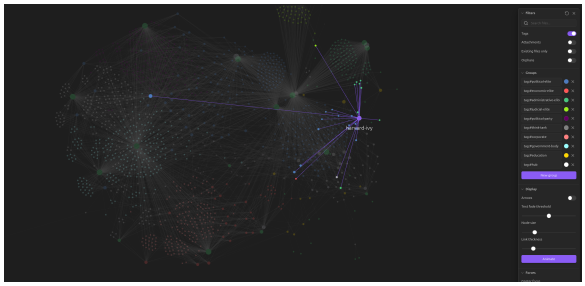


Figure 2: Detail of commissioner think-tank cluster. Commissioners (blue) connected to Atlantic Council, Trilateral Commission, and Friends of Europe (grey think-tank nodes).

4 Findings

The following findings should be read with appropriate caution. Our dataset covers 150 commissioners, 168 MEP leaders, 85 DGs, and 37 corporate leaders—adequate for descriptive statistics but modest by the standards of large-N social science. Sample error may be substantial, particularly for subgroup analyses (e.g., CJEU judges at 22/39 education coverage, corporate elites at 34/52). The more fine-grained qualitative claims—such as the apparent shift from *grandes écoles* to regional universities in French commissioner education—are based on small numbers of individuals and could be revised by additional data collection. We report them as suggestive patterns that warrant further investigation, not as definitive results.

The methodology surfaces several substantive findings:

4.1 Collapse of the French Elite Pipeline

French commissioners from Santer through Von der Leyen I were consistently products of the *grandes écoles*: Pascal Lamy (ENA + Sciences Po + HEC), Pierre Moscovici (ENA + Sciences Po), Jacques Barrot (Sciences Po). The current French commissioner, Stéphane Séjourné, attended the University of Poitiers—a mid-tier regional university with no elite pipeline connection. He is the first French commissioner since 1995 with no Paris education of any kind. Across all commissions, elite institution concentration has declined from 37% (Barroso I) to 8% (Von der Leyen II).

4.2 Elite Network Asymmetry

Commissioners share multiple transnational elite networks (Bilderberg Group: 10 commissioners, ECFR: 9, Trilateral Commission: 7, Friends of Europe: 12). MEP leaders, across 171 identified SDT-relevant individuals, cluster in political party structures and parliamentary bodies (European People's Party: 7 MEPs, Party of European Socialists: 5) rather than policy institutes. DGs and DDGs form a third distinct network centred on the College of Europe (6 DGs) and professional associations. Only 4 of 22 corporate leaders share commissioner network affiliations (Jean Lemierre and Oliver Bäte bridge the Trilateral Commission and Atlantic Council). Each elite group operates in a distinct network space—commissioners in policy institutes, MEPs in party structures, DGs in professional-bureaucratic associations, and corporate leaders in industry roundtables and financial networks.

4.3 EP-to-Commission Circulation

Only six of 150 commissioners held EP leadership positions (President, Vice President, or committee chair) before their Commission appointment. Circulation peaked at two per term under Barroso and Juncker, and has fallen to zero in the current Von der Leyen II term. When commissioners do have MEP backgrounds, they are typically national ministers who served brief MEP stints as a transit lounge between national power and Brussels—not career EP politicians who climbed the parliamentary ranks.

4.4 Elite School Specialisation

Table 1 shows institutions with six or more attendees, broken down by elite track.

The College of Europe is the most balanced pipeline, producing political, administrative, and judicial elites in roughly equal measure—a reflection of its explicit mission to train European civil servants. Sciences Po and UCL are heavily political tracks. École Polytechnique is the most specialised: five of its six attendees are corporate leaders, reflecting the French engineering *grandes écoles* to executive boardroom pipeline. The data suggests that elite education is not

Institution	Political	Admin.	Judicial	Corporate	Total
College of Europe	3	6	1	0	10
Sciences Po	8	0	1	0	9
Harvard	6	2	1	0	9
UCL	7	1	0	0	8
ULB	5	1	1	0	7
Sapienza Rome	6	0	0	1	7
Royal Malta	5	1	1	0	7
École Polytechnique	1	0	0	5	6
Oxford	3	1	0	1	5
Cambridge	2	1	0	0	3

Table 1: Elite school specialisation by track. Political = commissioners + MEPs; Admin. = Directors-General and deputies; Judicial = CJEU judges and advocates general; Corporate = transnational corporate leaders.

a single pipeline but a set of parallel tracks: College of Europe for the administrative elite, Sciences Po for the political elite, École Polytechnique for the economic elite.

Institutions with six or more attendees reveal distinct elite formation tracks. The College of Europe (10 attendees) is the most balanced pipeline, producing political, administrative, and judicial elites in roughly equal measure—a reflection of its explicit mission to train European civil servants. Sciences Po (9) and University College London (8) are heavily political tracks, feeding commissioners and MEPs almost exclusively. École Polytechnique (6) is the most specialised: five of its six attendees are corporate leaders, reflecting its role as the pipeline from French engineering *grandes écoles* to executive boardrooms. Harvard (9) feeds all tracks but are politically dominated. The data suggests that elite education is not a single pipeline but a set of parallel tracks: College of Europe for the administrative elite, Sciences Po for the political elite, École Polytechnique for the economic elite.

Only six of 150 commissioners held EP leadership positions (President, Vice President, or committee chair) before their Commission appointment. Circulation peaked at two per term under Barroso and Juncker, and has fallen to zero in the current Von der Leyen II term. When commissioners do have MEP backgrounds, they are typically national ministers who served brief MEP stints as a transit lounge between national power and Brussels.

5 Future Work

The corporate layer has been partially addressed: 52 leaders from 25 of 30 target companies now have education (65%) and organisation affiliation (87%) data. The remaining gap is board members for the last 5 companies (Volkswagen, Leonardo, Thales, Santander, ING), where Wikipedia page formats resisted automated extraction. CJEU coverage now stands at 23/39 judges and advocates general for education (59%), with the remaining 16 lacking either Wikipedia or Wikidata P69 data. DG/DDG organisation affiliations have been extracted for 67 of 85 officials (79%) from their CV PDFs.

Two structural extensions remain for future phases. First, the General Court judges—67 additional judicial officials identified during CJEU scraping whose bios were captured but not yet processed into entity records—would nearly triple the judicial dataset. Second, a formal overrepresentation metric based on Eurostat tertiary education statistics would quantify elite pipeline strength for specific institutions, allowing claims of "elite signalling" to be tested against population baselines rather than asserted.

6 Conclusion

This paper's own bibliography was verified using the methodology it describes. Each cited work was downloaded as a PDF, checked by `check_citations.py`—a script that extracts citation claims from the LaTeX source, searches the cited PDF text, and sends numbered phrases to an LLM for phrase-level verification with confidence scoring. At the time of writing, all citations pass the checker's verification threshold. The verification report is committed alongside this paper and can be regenerated by any reader with API access.

The key methodological claim of this paper is that LLM-assisted research can be rigorous if the LLM is treated as a hypothesis generator rather than an authority. The specific mechanisms—phrase-indexed manifests, mandatory provenance, cross-validation against source text, and contextualised LLM judging for entity resolution—are transferable to other domains. The case study demonstrates that this discipline yields not just auditable data but substantive findings that manual methods would require years to surface.

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